

VEERAVALLI SHANMUKHA SANTHOSH RAO

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Professional Summary

M.Tech Computer Engineering graduate candidate with hands-on experience in Machine Learning, Full Stack Development, Computer Vision, and Signal Processing. Developed AI-powered healthcare and mobile sensing applications using Python, FastAPI, TensorFlow, OpenCV, and XGBoost. Experienced in building end-to-end software solutions through internships and research projects. Passionate about developing scalable applications and leveraging data-driven technologies to solve real-world problems.

Technologies

Programming Languages: Python, Java, C, C# (Basic), C++ (Basic), JavaScript, Kotlin, Swift

Web Development: HTML5, CSS3, React.js, Node.js, Express.js

Mobile Development: Android Studio, Kotlin, Java, CameraX, ML Kit Face Detection, Swift, Xcode, AV-Foundation, Vision Framework, UIKit, Core Image, iOS Application Development

Tools: Git, GitHub, Linux (Basic), Streamlit, FastAPI, MATLAB, VS Code

Databases: MongoDB, SQL

Machine Learning & AI: TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy

Computer Vision & Signal Processing: OpenCV, rPPG, VPG, Digital Signal Processing (DSP)

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Linux/UNIX Fundamentals, Socket Programming, DBMS, Software Engineering, Software Development Life Cycle (SDLC), REST APIs, Machine Learning, Full Stack Development

Networking: Socket Programming, TCP/IP Fundamentals

Soft Skills: Problem Solving, Communication, Team Collaboration, Time Management, Work Ethic

Productivity Tools: Microsoft Word, Microsoft Excel, Microsoft PowerPoint

Relevant Coursework

- Data Structures & Algorithms
- Operating Systems
- Computer Networks
- Database Management Systems
- Software Engineering
- Object-Oriented Programming

Education

PARUL UNIVERSITY

M.Tech in Computer Engineering - [GPA: 7.41/10.0]

Aug 2024– May 2026

R.V.R. J.C. COLLEGE OF ENGINEERING

B.Tech in Computer Science and Engineering - [GPA: 6.93/10.0]

Jan 2021– May 2024

NARAYANA JUNIOR COLLEGE

Board of Intermediate Education (AP) - [GPA: 8.25/10]

Jun 2018 – Mar 2020

SRI VENKATESWARA BALA KUTEER

Central Board of Secondary Education (CBSE) - [GPA: 5.8/10]

Apr 2017 – May 2018

Experience

SLASH MARK

Jan 2024 - Apr 2024

Full Stack Web Development Intern

- Developed responsive web applications using frontend and backend technologies.
- Integrated databases and APIs to support dynamic application functionality.
- Collaborated on debugging, testing, and deployment activities to improve

application performance.

- Applied software development best practices to enhance user experience and maintainability.

YOSHOPS

May 2023 - Jul 2023

Data Science Intern

- Performed data preprocessing, analysis, and visualization on business datasets.
- Assisted in developing billing and reporting solutions to improve operational workflows.
- Utilized Python-based tools for data management and business analytics.
- Generated insights that supported data-driven decision-making processes.

INDIAN SERVERS

Apr 2022 - Jun 2022

Software Trainee

- Built Python-based machine learning applications using industry-standard libraries.
- Implemented data preprocessing and feature engineering techniques on real-world datasets.
- Explored supervised learning algorithms and model evaluation methodologies.
- Strengthened software development and problem-solving skills through practical projects.

Projects

Real-Time Non-Contact Pulse Rate and SpO2 Monitoring Using Mobile Cameras



March 2026

- Developed a cross-platform healthcare monitoring system for Android, iOS, and Web platforms using remote photoplethysmography (rPPG) technology.
- Developed Android applications using Kotlin, CameraX, ML Kit Face Detection, and OpenCV for real-time physiological signal acquisition and analysis.
- Developed iOS application using Swift, AVFoundation, Vision Framework, and Core Image for real-time pulse monitoring and physiological signal extraction.
- Designed face detection, ROI extraction, and image-processing pipelines to capture facial blood-volume pulse signals from smartphone cameras.
- Implemented signal-processing and machine-learning techniques for pulse rate estimation, SpO trend analysis, and cardiac rhythm monitoring.
- Developed backend services using FastAPI and integrated OpenCV, TensorFlow, and XGBoost for data processing and inference.
- Enabled contactless health monitoring through real-time camera analysis across mobile and web environments.
- **Tech Stack:** Python, Kotlin, Swift, FastAPI, OpenCV, TensorFlow, XGBoost, Android Studio, CameraX, ML Kit Face Detection, AVFoundation, Vision Framework, HTML, CSS, JavaScript, Signal Processing, FFT, Bandpass Filtering, rPPG, VPG

Potential Application of Using Smartphone Sensor for Estimating Air Temperature: Experimental Study



May 2024

- Developed a mobile application to estimate air temperature using battery sensor data.
- Implemented regression models for prediction and improved estimation accuracy.
- Utilized Python and ML techniques for experimental validation.
- **Tech Stack:** Python, Machine Learning, Scikit-learn, Pandas, NumPy, Android Sensors, Data Analytics, Android Studio

An End to End Chatbot using Python

July 2023

- Developed a conversational chatbot using Python and Streamlit.
- Integrated web scraping and data processing using pandas and numpy.
- Automated responses to handle user queries without human intervention.
- **Tech Stack:** Python, Streamlit, Pandas, NumPy, Web Scraping, NLP

Face Mask Detection

June 2022

- Built a CNN-based model using TensorFlow and Keras for mask detection.
- Achieved high accuracy in real-time image classification.
- Applied computer vision techniques for detection and analysis.
- **Tech Stack:** Python, TensorFlow, Keras, OpenCV, NumPy, CNN



A Review of Monitoring Heart Rate and Cardiac Rhythm by video photoplethysmography on Mobile Devices

November 2025

- Conducted a comprehensive review of remote photoplethysmography (rPPG) and video photoplethysmography (VPG) frameworks deployed on mobile devices.
- Analyzed diverse machine learning architectures and digital signal processing pipelines optimized for real-time cardiac rhythm extraction.
- Evaluated the challenges of camera-based, non-contact physiological monitoring, presenting critical benchmarks for future mobile health deployments.
- **Tech Stack:** Research Work, Remote Photoplethysmography (rPPG), Video Photoplethysmography (VPG), Signal Processing, Machine Learning, Python, OpenCV, FFT Analysis

Publications

- Veeravalli S. S. Rao, V. Kaushik, D. Swami, and K. R. Jadav, "A Review of Monitoring Heart Rate and Cardiac Rhythm by Video Photoplethysmography on Mobile Devices," *Proceedings of the 2nd International Conference on Recent Advancement and Modernization in Sustainable Intelligent Technologies & Applications (RAMSITA 2026)*, Atlantis Press, 2026. DOI: 10.2991/978-94-6239-678-4_17.

Certifications

Hands-on Workshop on GenAI: Building RAG and Agentic AI Application - 8th Parul University International Conference on Engineering and Technology PiCET 2026	2026
Microsoft Azure Machine Learning Fundamentals - LinkedIn Learning	2024
Introduction to Industry 4.0 and Industrial Internet of Things (Elite) - NPTEL	2023
HTML Certificate - IIT Bombay	2023
Image Processing Onramp - MATHWORKS (MATLAB)	2022
AWS Academy Cloud Foundations - AWS Academy	2022
Principles of Management (Elite) - NPTEL	2022
Introduction to Programming using Python - Microsoft	2022
Programming in Java (Elite) - NPTEL	2022
The joy of computing using Python (Elite Silver) - NPTEL	2021

Research Professional Development

- Conducted independent research in remote photoplethysmography (rPPG), video photoplethysmography (VPG), and mobile health monitoring systems.
- Published research in the healthcare AI and signal processing domain.
- Earned certifications in Machine Learning, Cloud Computing, Programming, and Image Processing from recognized platforms.
- Continuously explored emerging technologies in Artificial Intelligence, Full Stack Development, and Software Engineering.